

9.1 Current & Potential Difference

Question Paper

Course	CIEA Level Physics
Section	9. Electricity
Topic	9.1 Current & Potential Difference
Difficulty	Medium

Time allowed: 50

Score: /34

Percentage: /100

Question 1a

The power P dissipated in a resistor of resistance R is measured for a range of values of the potential difference V across it.

The results are shown in the table in Fig. 1.1.

$V /$	$V^2 /$	$P /$
1.00	1.0	0.214
1.71	2.9	0.58
2.25		1.011
2.67		1.43
3.00	9.0	1.80
3.27	10.7	2.176
3.50	12.3	2.43

Fig. 1.1.

For the table in Fig. 1.1

i)
Add the correct units.

[1]

ii)
Complete the missing data.

[1]

iii)
Identify errors in the recorded the data and correct them.

[1]

[3 marks]

Question 1b

Complete the graph of the data on Fig. 1.2. below.

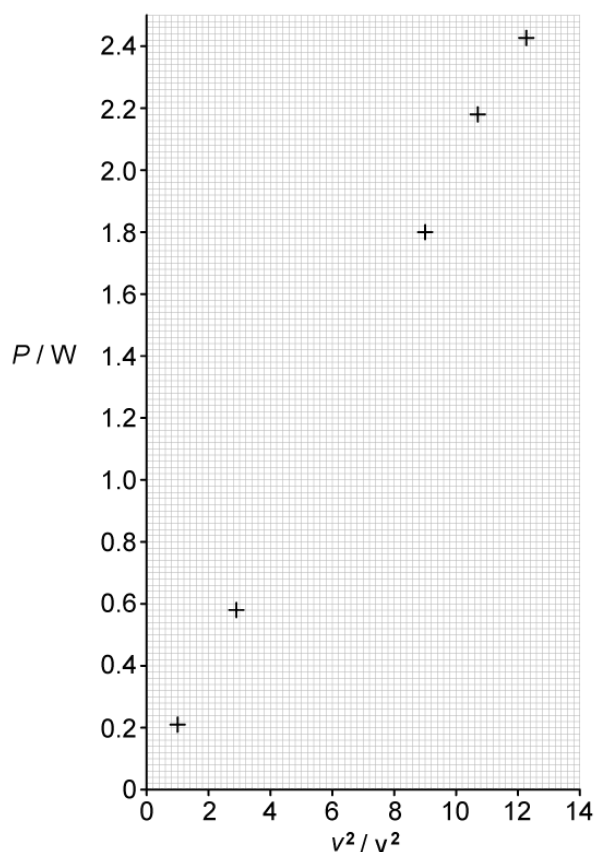


Fig. 1.2.

i)
Plot the two missing points **and** complete the graph.

[3]

ii)
Determine the gradient of the graph.

[3]

iii)
Use the gradient of the graph to obtain a value for R .

[1]

[7 marks]

Question 1c

The rest of this question is based on the data in the table Fig. 1.1 in part (a).

i)

Determine the value of R when $V = 3.50 \text{ V}$.

[1]

ii)

The uncertainty in V is $\pm 0.01 \text{ V}$.

The uncertainty in P is $\pm 0.05 \text{ W}$.

Calculate the percentage uncertainty in the value of R calculated in part (i).

[3]

iii)

Hence calculate the uncertainty in the value of R .

[1]

iv)

State and explain whether the value of R you calculated in part (i) is consistent with the value of R you determined from the gradient in part (b).

[2]

[7 marks]

Question 2a

The smallest conductor within a computer processing chip can be represented as a rectangular block that is one atom high, four atoms wide and twenty atoms long. One such block is shown in Fig. 1.1.

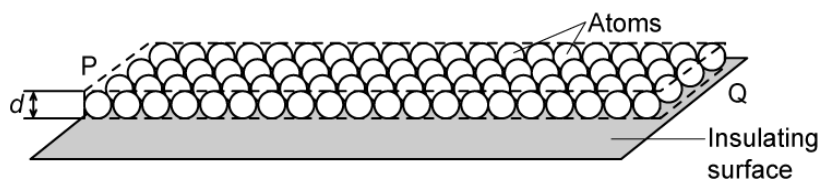


Fig. 1.1.

The block is made from phosphorus atoms of diameter $d = 3.8 \times 10^{-10}$ m. The atoms are deposited on an insulating surface. This ensures that the atoms touch each other.

The processing chip has a resistance, measured from **P** to **Q**, of $220 \, \Omega$.

i)
Define current.

[1]

ii)
Describe for this processing chip the properties which will affect the current.

[3]

[4 marks]

Question 2b

Calculate the number density of free electrons within the block.

Assume that each phosphorus atom contributes one free electron.

[1 mark]

Question 2c

Calculate the current between P and Q when the mean drift velocity of free electrons in the block is $1.9 \times 10^{-5} \text{ ms}^{-1}$.

[3 marks]

Question 2d

There are about 10^9 of these tiny conductors in a single chip each carrying the current calculated in part (c).

Estimate the total power dissipated in these conductors in a single chip.

[3 marks]

Question 3a

A power source is connected across a component. The current in the component is 40 mA.

Calculate the total charge that flows past a point in the component in 3 minutes.

[1 mark]

Question 3b

Calculate the number of electron charge carriers passing the same point in the conductor in this time.

[1 mark]

Question 3c

8.6 J of energy are transferred to the conductor in this time.

i)

Calculate the potential difference across the component.

[2]

ii)

Calculate the resistance of the conductor.

[2]

[4 marks]



Model Answers: Medium

1a

a) A correctly completed table looks like:

V/V	V^2/V^2	P/W
1.00	1.0	0.21
1.71	2.9	0.58
2.25	5.1	1.01
2.67	7.1	1.43
3.00	9.0	1.80
3.27	10.7	2.18
3.50	12.3	2.43

i) Add correct units.

- All three units correct; [1 mark]

ii) Complete the missing data.

Square the corresponding value of V :

- Two missing data correct; [1 mark]

iii) Identify errors in the recorded the data and correct them:

- Three values of power rounded to two decimal places; [1 mark]

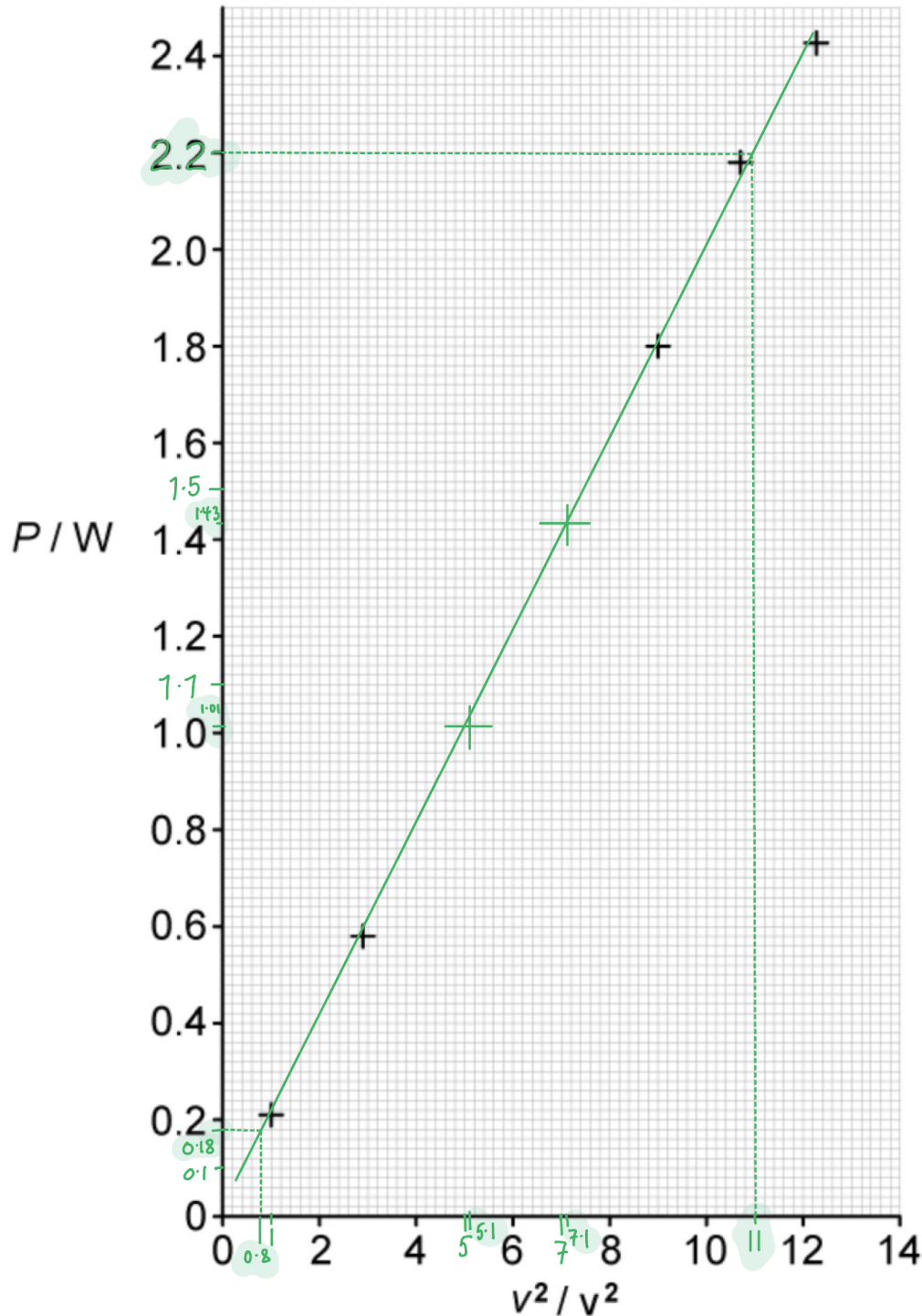
[Total: 3 marks]

1b

b) Complete the graph of the data on Fig. 1.2.

i) Plot the two missing points and complete the graph

A correct graph looks like:



- Plot at (5.1, 1.01) correct to within half a small square; [1 mark]
- Plot at (7.1, 1.43) correct to within half a small square; [1 mark]
- *Line of best fit*: thin, straight line (drawn with ruler) with approximately same

number of plots on each side of the line; [1 mark]

ii) Determine the gradient of the graph

- Clear indication marked on the graph that the large triangle method has been used (occupying at least two thirds of the line); [1 mark]

Gradient calculation

- $\text{gradient}, m = \frac{\Delta y}{\Delta x}$
- $m = \frac{2.2 - 0.18}{11 - 0.8}$ [1 mark]
- $m = 0.198 \text{ (W V}^{-2}\text{)}$ (answer in the range 0.190 - 0.120, to 2 or 3 sig figs) [1 mark]

iii) Use the gradient of the graph to obtain a value for R

- Since gradient represents P/V^2 and $R = V^2/P$, then R is the reciprocal of the gradient
- $R = \frac{1}{0.198} = 5.05 = 5.1 \Omega$ [1 mark]

[Total: 7 marks]

We have left the 'working out' annotations on our graph to demonstrate how important these are. Once you are finished with a graph question it should look like ours. By marking the scale we have made sure to get our plots correct. It's also worth trying the line of best fit a second time if you aren't sure about the first one.

Your teacher wants you to work in pencil so that you can erase mistakes. This is very, very good advice, which you should follow. However, if you go over your final line in pen, and then rub out working or mistakes, your *examiner* will not mind. They are looking for accuracy.

1c

c) The rest of this question is based on the data in the table Fig. 1.1 in part (a)

i) Determine the value of R when $V = 3.50 \text{ V}$

Use the data table to determine P when $V = 3.50 \text{ V}$

- $P = 2.43 \text{ W}$

Use the correct equation and substitute:

- $R = \frac{V^2}{P} = \frac{3.5^2}{2.43} = 5.04$
- $R = 5.0 \text{ or } 5.04 \Omega \text{ or } 5.06 \Omega$ [1 mark]

ii) Calculate the percentage uncertainty in the value of R calculated in part (i):

List the known quantities:

- Uncertainty in $V = \pm 0.01 \text{ V}$
- Value of $V = 3.5 \text{ V}$
- Uncertainty in $P = \pm 0.05 \text{ W}$
- Value of $P = 2.43 \text{ W}$

Determine the percentage uncertainties in V and P :

All marks are for answers to 2 or 3 sig figs **only**

- Percentage uncertainty in V , $\%_V = \left(\frac{0.01}{3.5} \right) \times 100 = 0.29 \%$ [1 mark]
- Percentage uncertainty in P , $\%_P = \left(\frac{0.05}{2.43} \right) \times 100 = 2.1 \%$ [1 mark]

Write an expression to combine the percentage uncertainties in V and P and substitute:

- Percentage uncertainty in R , $\%_R = (2 \times \%_V) + \%_P$
- $\%_R = (2 \times 0.29) + 2.1 = 2.7 \%$ [1 mark]

iii) Hence calculate the uncertainty in the value of R :

- Absolute uncertainty = percentage uncertainty $\times$ measured (or calculated) value

- Uncertainty of R , $U_R = 0.027 \times 5.04 = 0.136 = \pm 0.1$ **or** $\pm 0.14 \Omega$ [1 mark]

iv) State and explain whether the value of R you calculated in part (i) is consistent with the value of R you determined from the gradient in part (b):

Use the calculated value and uncertainty to establish a range of values:

- R from calculation = $5.04 \pm 0.14 \Omega$ **or** $5.06 \pm 0.14 \Omega$
- Therefore, $4.90 \Omega \leq R \leq 5.18 \Omega$; [1 mark]

Compare to the value derived from the graph:

- R from graph = 5.05Ω
- Therefore the value from the graph falls within the calculated range so they are consistent; [1 mark]

[Total: 7 marks]

2a

a)

i) Define current:

- A flow of charge carriers; [1 mark]

ii) Describe for this processing chip the properties which will affect the current:

Any **three** from:

- $I = Anvq$ (any arrangement); [1 mark]
- Charge per carrier / electron = $1.9 \times 10^{-19} \text{ C}$; [1 mark]
- Number of electrons / charge density is constant; [1 mark]
- Current increases for
 - increased potential difference / voltage across the component; [1 mark]
 - increased cross-sectional area; [1 mark]

[Total: 4 marks]

2b

b) Calculate the number density of free electrons within the block:

Since each atom contributes one electron, calculate the number of atoms per unit volume:

- $n = \frac{1}{d^3} = \frac{1}{(3.8 \times 10^{-10})^3} = 1.8 \times 10^{28} \text{ [1 mark]}$

[Total: 1 mark]

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- Number of electrons / charge density is constant; [1 mark]
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[Total: 1 mark]

2c

c) Calculate the current between P and Q when the mean drift velocity of free electrons in the block is $1.9 \times 10^{-5} \text{ ms}^{-1}$

List the known quantities:

- Drift velocity of free electrons, $v = 1.9 \times 10^{-5} \text{ ms}^{-1}$

- Charge on an electron, $q = -1.6 \times 10^{-19} \text{ C}$ (from the data sheet)
- Number density of electrons, $n = 1.8 \times 10^{28}$ (from part b)
- Diameter of atom, $d = 3.8 \times 10^{-10} \text{ m}$ (from part a)

Determine the cross-sectional area of the conductor:

- The current flowing from P to Q moves through a surface area 4 atoms long by 1 atom wide
- $A = 4 \times (3.8 \times 10^{-10})^2 = 5.776 \times 10^{-19} \text{ m}^2$ [1 mark]

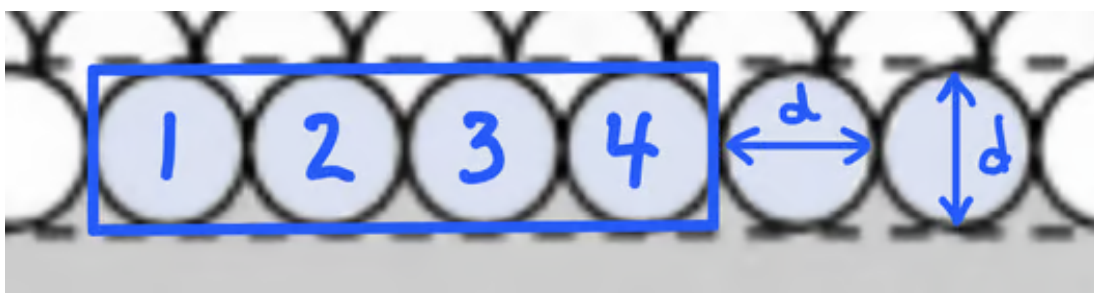
Use the equation relating current and drift velocity:

- $I = Anvq$
- $I = (5.776 \times 10^{-19}) \times (1.8 \times 10^{28}) \times (1.9 \times 10^{-5}) \times (1.6 \times 10^{-19})$ [1 mark]
- $I = 3.2 \times 10^{-14} \text{ A}$ [1 mark]

[Total: 3 marks]

Using the diameter of a sphere to calculate the surface area of a cuboid may seem odd. It is an approximation - what we call a 'best estimate' - but using this kind of informed shortcut is an important aspect to calculations in physics.

The sketch below shows how the idea works. You can see that although there are gaps, they are small, so the estimation is a valid one.



You can also see that the cross-sectional area of the conductor is:

$$(4 \times \text{atom}) \times (1 \times \text{atom})$$

2d

d) Estimate the total power dissipated in these conductors in a single chip:

List the known quantities:

- Current, $I = 3.2 \times 10^{-14} \text{ A}$
- Resistance, $R = 220 \Omega$ (from part a)
- Number of components in series $= 10^9$

Determine the total resistance:

- $R = 220 \times 10^9 = 2.2 \times 10^{11} \Omega$ [1 mark]

Use the equation for power and calculate:

- $P = I^2 R = (3.2 \times 10^{-14})^2 \times (2.2 \times 10^{11})$ [1 mark]
- $P = 2.3 \times 10^{-16} \text{ W}$ [1 mark]

[Total: 3 marks]

The resistors are in **series**, so the total resistance will be the **sum** of all the resistors.

$$R_T = R + R + R + R \dots = nR$$

3a

a) Calculate the total charge that flows past a point in the component in 3 minutes:

List the known quantities:

- Current, $I = 40 \text{ mA} = 40 \times 10^{-3} \text{ A}$
- Time, $t = 3 \text{ minutes} = 180 \text{ s}$

Use the equation for charge and calculate:

- $Q = It = (40 \times 10^{-3}) \times 180 = 7.2 \text{ C}$ [1 mark]

[Total: 1 mark]

3b

b) Calculate the number of electron charge carriers passing the same point in the conductor in this time:

List the known quantities:

- Total charge, $Q = 7.2 \text{ C}$ (from part a)
- Charge per electron, $q = -1.6 \times 10^{-19}$ (from the data booklet)

Calculate the total number of electrons:

$$\bullet \quad n = \frac{Q}{q} = \frac{7.2}{(1.6 \times 10^{-19})} = 4.5 \times 10^{19} \text{ [1 mark]}$$

[Total: 1 mark]

3c

c)

i) Calculate the potential difference across the component:

List the known quantities:

- Energy, $W = 8.6 \text{ J}$
- Charge, $Q = 7.2 \text{ C}$

Use the equation defining potential difference:

$$\bullet \quad V = \frac{W}{Q} = \frac{8.6}{7.2} \text{ [1 mark]}$$

$$\bullet \quad V = 1.2 \text{ V (2 s.f.) [1 mark]}$$

ii) Calculate the resistance of the conductor:

List the known quantities:

- Potential difference, $V = 1.2 \text{ V}$
- Current, $I = 40 \times 10^{-3} \text{ A}$ (from part a)

Use the equation for resistance:

$$R = \frac{V}{I} = \frac{1.2}{40 \times 10^{-3}} = 3.0 \text{ } \Omega$$

- $R = \frac{V}{I} = \frac{1.2}{(40 \times 10^{-3})}$ [1 mark]

- $R = 30 \, \Omega$ [1 mark].

[Total: 4 marks]

